QP CODE: **Reg. No. : .....**
Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG3DSEPHY203– INTRODUCTION TO OPTICS

Duration: 1.5 hrs Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions
Answer any **10** questions
Each question carries **2** Mark

1.	Foucault's and Michelson's experimental measurements of the speed of light disproved which prediction of Newton's corpuscular theory?	[A][1]
2.	Estimate the frequency of light corresponding to a wavelength of 500 nm.	[U][2]
3.	Calculate the equivalent optical path length if light propagates through a glass block of length 1 m and refractive index = 1.5	[A][3]
4.	What happens to the path of a light ray entering from one medium to another if both medium have same refractive indices?	[An][3]
5.	Explain paraxial approximation in optics.	[U][4]
6.	Which two parameters are required to describe a ray with respect to optical axis?	[U][4]
7.	Which are the two operators used to describe the propagation of a ray through an optical system in the matrix method?	[U][4]
8.	Compare the shapes of a primary wavefront very near to a point source of light and that of a secondary wavefront far away from the source	[A][4]
9.	What is meant by a thin lens?	[U][4]
10.	Describe the role of entrance and exit pupils in light transmission.	[U][6]
11.	Two lenses have the same focal length, but one has a higher f-number than the other. Which one will produce a brighter image under identical lighting conditions?	[An][6]
12.	Distinguish between the objective and the eyepiece in an optical system.	[An][6]
13.	What are the two factors that determine the overall angular magnification of a compound microscope?	[U][6]
14.	How can spherical aberrations be avoided in a reflecting telescope?	[A][6]

[2×10 = 20]

Part B

Short Essay Type Questions
Answer any **6** questions
Each question carries **5** Mark

15. Discuss Maxwell's electromagnetic theory and name the experiment that confirmed it. [U][1]

16. Using Fermat's principle of least time, derive Snell's law for refraction at the boundary between two media with different refractive indices. [A][3]

17. A ray of light travels inside a glass block of refractive index 1.67 and strikes the interface with an unknown liquid. It is observed that total internal reflection occurs when the angle of incidence exceeds the critical angle 53 degrees. (a) Calculate the refractive index of the liquid. (b) Identify the liquid based on your result. [An][3]

18. Derive the refraction matrix at a spherical surface and express it in terms of the power of the surface. [A][4]

19. The radii of curvature of the two surfaces of a double convex lens are +15 cm and −25cm, and its focal length is found to be 30cm in air. Determine the refractive index of the material of the lens. [A][4]

20. Based on Huygens' principle, discuss the reflection of a plane wavefront at a plane reflecting surface, and derive the relation between the angle of incidence and the angle of reflection. [A][5]

21. Draw the ray diagram of a simple refracting astronomical telescope made using two convex lenses and derive the expression for its magnifying power. Interpret the sign (positive or negative) of the magnification obtained. [An][6]

22. A thin convex lens has focal lengths of 98 cm for blue light and 102 cm for red light. Calculate the dispersive power of the material and the mean focal length of the lens. [A][6]

[5×6 = 30]